

Ravi Chandran Reddy Kallem

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Profile

Versatile Data Engineer with 4+ years of experience in building **scalable data pipelines**, cloud architectures, and software solutions. Proficient in Python, Java, Scala, SQL, and Big Data tools like **HDFS, Kafka, and Hive**. Skilled in AWS services (EC2, S3, Redshift, EMR), Terraform, and modern databases including PostgreSQL and MongoDB. Experienced in data visualization with Power BI, Airflow, and QuickSight, and adept at DevOps practices using CI/CD pipelines, Docker, and Kubernetes. Known for optimizing workflows, automating processes, and delivering impactful solutions in fast-paced environments.

Technical Skills

- Programming Languages:** Python, Java, Scala, SQL
- Big Data Frameworks & Tools:** HDFS, MapReduce, Oozie, Hive, Pig, Sqoop, HBase, Kafka
- Cloud Platforms & Infrastructure:** AWS, Terraform, Snowflake
- AWS Services:** EC2, S3, IAM, Redshift, Lambda, SNS, EMR, Application Load Balancer (ALB)
- Databases:** Amazon Redshift, PostgreSQL, MySQL, DynamoDB, HDFS, HBase, Cassandra, MongoDB
- Data Visualization & Workflow Tools:** Airflow, Amazon QuickSight, Power BI, SQL Workbench, IntelliJ IDEA
- DevOps & Automation Tools:** CI/CD Pipelines, Docker, Kubernetes
- Additional Technologies:** GraphQL, ReactJS, Git, GitHub

Work Experience

Ameriprise Financial | Hadoop Developer

Feb 2024 – Present

- Designed and implemented scalable data processing pipelines** using Hadoop-HDFS, Hive, and Pig, enabling the processing of multi-terabyte datasets with a 30% improvement in processing efficiency.
- Automated ETL workflows** by integrating Oozie with Sqoop for seamless data ingestion from relational databases like PostgreSQL and MySQL into HDFS, reducing manual effort by 40%.
- Optimized distributed data storage and retrieval processes** by leveraging HBase and Cassandra, achieving a 25% reduction in query response times for high-velocity data streams.
- Developed real-time data streaming solutions** using Kafka, ensuring reliable ingestion and processing of large-scale data for time-sensitive business operations.
- Streamlined DevOps processes** by building CI/CD pipelines with Docker and Kubernetes, enabling faster deployments and improving system reliability.
- Enhanced monitoring and workflow automation** with Apache Airflow, improving task orchestration efficiency and ensuring on-time data delivery for critical business reporting.

Apple, Inc| Big Data Engineer

Mar 2020 – Jun 2022

- Improved ETL pipeline efficiency by 30% through the integration of Spark-Scala and Hadoop-HDFS**, enhancing geocoding accuracy for location services and ensuring reliable mapping in global-scale applications.
- Automated geometry matching for large-scale geospatial datasets** using 3D vector data, Spark, and HDFS, achieving a 25% increase in mapping precision and reducing processing time in the ETL pipeline.
- Optimized big data workflows by automating Pig, Hive, and Spark task orchestration** with custom shell scripts, boosting data processing speed by 40% and cutting operational overhead by 15%.
- Maintained a 90% code coverage standard by implementing unit testing and RDD-based testing** using JUnit and RDDAssert, ensuring production pipeline reliability and minimizing post-deployment issues.
- Enhanced software reliability and version control** by enforcing rigorous code review standards and leveraging Git for collaborative development and robust versioning.

Infosys, Ltd. | Software Engineer

May 2019 – Mar 2020

- Designed and deployed a large-scale AWS Data Lake architecture** for a Fortune 500 client, leveraging AWS Glue, S3, and Lambda to process over 10TB of geospatial data, improving data ingestion and processing speeds by 40%.
- Implemented a high-performance analytics framework** using AWS Redshift and Redshift Spectrum, enabling rapid analysis of 10TB+ datasets and reducing time-to-insight by 40%.
- Automated CI/CD pipelines using GitHub**, AWS CodePipeline, and Selenium, cutting deployment times for data workflows from 3 hours to 30 minutes and enhancing system reliability.
- Developed scalable APIs for cloud resource management** using Python, Flask, and AWS CDK, supporting 5000+ multi-region data processing jobs and streamlining operations for global teams.
- Enhanced data flow efficiency by 25% through robust ETL pipeline development** with AWS EMR and data modeling, collaborating with cross-functional teams to optimize business processes.
- Built a Skill Gap analysis dashboard for 50K+ employees**, analyzing over 1TB of data using HDFS, Hive, and Sqoop, providing actionable insights for HR to address training needs.

Education

University of Texas at Arlington, TX, USA

May 2024

Masters in Computer Science and Mathematics

Osmania University, Hyderabad, India

May 2019

Bachelors in Computer Science and Engineering